

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-7

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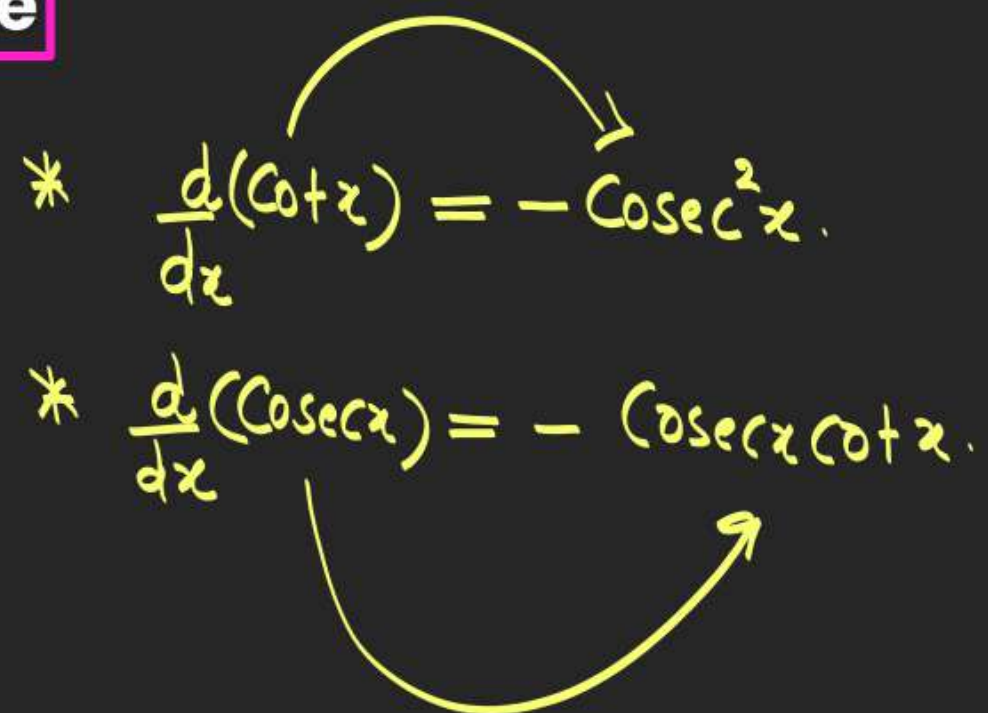
TOPICS TO BE COVERED

1

Exercise Discussion 12.2

2

Miscellaneous Exercise

$$\begin{aligned}
 * \quad \frac{d(\cot x)}{dx} &= -\operatorname{cosec}^2 x. \\
 * \quad \frac{d(\operatorname{cosec} x)}{dx} &= -\operatorname{cosec} x \cot x.
 \end{aligned}$$


QUESTION (NCERT EXAMPLER):

If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then $\frac{dy}{dx}$ at $x = 1$ is

$$\frac{1}{x^{1/2}} = x^{-1/2}$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

1 1 ✗

2 $1/2$ ✗

3 $1/\sqrt{2}$ ✗

4 0 ✓

(Sol) Given that $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

$$\Rightarrow y = x^{1/2} + x^{-1/2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{1/2-1} - \frac{1}{2} x^{-1/2-1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-1/2} - \frac{1}{2} x^{-3/2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2x^{1/2}} - \frac{1}{2x^{3/2}} = \frac{1}{2} \left(\frac{1}{\sqrt{x}} - \frac{1}{x\sqrt{x}} \right)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = \frac{1}{2} (1-1) = 0$$

QUESTION (NCERT EXAMPLER):

If $f(x) = \frac{x-4}{2\sqrt{x}}$ then $f'(1)$ is

1 $\frac{5}{4}$

2 $\frac{4}{5}$

3 1

4 0

(Sol) Given that $f(x) = \frac{x-4}{2\sqrt{x}}$

$$\Rightarrow f'(x) = \frac{1}{2} \left\{ \frac{\sqrt{x} \frac{d}{dx}(x-4) - (x-4) \frac{d}{dx} \sqrt{x}}{\cancel{\sqrt{x}}^2} \right\}$$

$$\Rightarrow f'(x) = \frac{1}{2} \left\{ \frac{\sqrt{x} (1-0) - \left(\frac{x-4}{2\sqrt{x}} \right)}{x} \right\}$$

$$\Rightarrow f'(x) = \frac{1}{2} \left\{ \frac{\cancel{2\sqrt{x}}^2 - x + 4}{2\sqrt{x}} \times \frac{1}{x} \right\}$$

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\Rightarrow f'(x) = \frac{x+4}{4x\sqrt{x}}$$

$$\text{So, } f'(1) = \frac{1+4}{4 \times 1} = \frac{5}{4}$$

$$\Rightarrow f'(1) = \frac{5}{4}$$

QUESTION (NCERT EXAMPLER):

If $y = \frac{1 + \frac{1}{x^2}}{1 - \frac{1}{x^2}}$, then $\frac{dy}{dx}$ is

1 $\frac{-4x}{(x^2 - 1)^2}$

2 $\frac{-4x}{x^2 - 1}$

3 $\frac{1 - x^2}{4x}$

4 $\frac{4x}{1 - x^2}$

(Sol) Given that $y = \frac{\left(1 + \frac{1}{x^2}\right)}{\left(1 - \frac{1}{x^2}\right)}$

$\Rightarrow y = \frac{\left(\frac{x^2+1}{x^2}\right)}{\left(\frac{x^2-1}{x^2}\right)}$

$\Rightarrow y = \frac{x^2+1}{x^2-1}$

So $\frac{dy}{dx} = \frac{(x^2-1) \frac{d(x^2+1)}{dx} - (x^2+1) \frac{d(x^2-1)}{dx}}{(x^2-1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{2x(x^2-1) - 2x(x^2+1)}{(x^2-1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{\cancel{2x^3} - 2x - \cancel{2x^3} - 2x}{(x^2-1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{-4x}{(x^2-1)^2}$

QUESTION (NCERT EXAMPLER):

If $f(x) = 1 + x + \frac{x^2}{2} + \dots + \frac{x^{100}}{100}$, then $f'(1)$ is equal to

- 1 $1/100$ ✗
- 2 100 ✓✓
- 3 Does not exist ✗
- 4 0 ✗

(Sol) Given that

$$f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{100}}{100}$$

$$\Rightarrow f'(x) = 0 + 1 + \cancel{\frac{2x}{2}} + \cancel{\frac{3x^2}{3}} + \dots + \cancel{\frac{100x^{99}}{100}}$$

$$\Rightarrow f'(x) = 1 + x + x^2 + x^3 + \dots + x^{99}$$

$$\Rightarrow f'(1) = 1 + 1 + 1^2 + 1^3 + \dots + (1)^{99}$$

$$\Rightarrow f'(1) = 1 + 1 + 1 + \dots + 1$$

$$\Rightarrow \boxed{f'(1) = 100}$$

QUESTION (NCERT EXAMPLER):

If $y = \frac{\sin(x + 9)}{\cos x}$ then $\frac{dy}{dx}$ at $x = 0$ is

1 ✓ $\cos 9$

2 ✗ $\sin 9$

3 ✗ 0

4 ✗ 1

(Sol) Given that $y = \frac{\sin(x+9)}{\cos x}$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x \frac{d}{dx} \sin(x+9) - \sin(x+9) \frac{d}{dx} (\cos x)}{(\cos x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x \cos(x+9) + \sin(x+9) \sin x}{(\cos x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos(\cancel{x} - \cancel{x} - 9)}{(\cos x)^2} = \frac{\cos(-9)}{(\cos x)^2}$$

$$\frac{d}{dx}(\sin(x+9)) = \cos(x+9) \times \frac{d}{dx}(x+9) = \cos(x+9)$$

$$\frac{d}{dx}(\sin(2x^2+x)) = \cos(2x^2+x) \times \frac{d}{dx}(2x^2+x) \text{ (chain rule)} \\ = (4x+1) \cos(2x^2+x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos 9}{(\cos x)^2}$$

$$\left(\frac{dy}{dx}\right)_{x=0} = \frac{\cos 9}{(\cos 0)^2} = \frac{\cos 9}{1^2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{x=0} = \cos 9$$

QUESTION (NCERT EXAMPLER):

If $f(x) = \frac{x^n - a^n}{x - a}$ for some constant ' a ', then $f'(a)$ is

Homework.

1

1 ✗

2

0 ✗

3

Does not exist ✓

4

$1/2$ ✗

EXERCISE – 12.2

QUESTION

Find the derivative of $x^2 - 2$ at $x = 10$.

$$\begin{aligned} &2x - 0 \\ &= 2x \bigg|_{x=10} = 20. \end{aligned}$$

QUESTION

Find the derivative of x at $x = 1$.

hp/w

QUESTION

Find the derivative of $99x$ at $x = 100$.

h/w

$$99x \longrightarrow 99.$$

QUESTION

(Important Question)

Find the derivative of the following functions from first principle.

(i) $x^3 - 27$ H/W

✓(ii) $(x-1)(x-2)$

(iii) $\frac{1}{x^2}$ H/W

✓(iv) $\frac{x+1}{x-1}$

(Sol) Suppose that $f(x) = \frac{x+1}{x-1}$ be the given

function. Then by first principle, we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left(\frac{\frac{x+h+1}{x+h-1} - \frac{x+1}{x-1}}{h} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left(\frac{(x-1)(x+h+1) - (x+1)(x+h-1)}{h(x+h-1)(x-1)} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{x^2} + \cancel{x}h + \cancel{x} - \cancel{x} - \cancel{h} - 1 - \cancel{x^2} - \cancel{x}h + \cancel{x} - \cancel{x} - \cancel{h} + 1}{h(x+h-1)(x-1)}$$

$$\Rightarrow f(x) = \lim_{h \rightarrow 0} \frac{-2}{(x+h-1)(x-1)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-2}{(x+h-1)(x-1)}$$

$$\Rightarrow f'(x) = \frac{-2}{(x+0-1)(x-1)}$$

$$\Rightarrow f'(x) = \frac{-2}{(x-1)^2} \text{ Hence the result.}$$

$$2x-3$$

(Sol)(ii) Suppose that $f(x) = (x-1)(x-2)$ be the given function.

$$\Rightarrow f(x) = x^2 - 3x + 2.$$

By first principle, we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - x^2 + 3x - 2}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2hx - 3x - 3h - x^2 + 3x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left(\frac{h^2 + 2hx - 3h}{h} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{h(h + 2x - 3)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} (h + 2x - 3)$$

$$\Rightarrow f'(x) = 2x - 3.$$

Hence the result.

QUESTION

For the function

$$f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1. \text{ Prove that } f'(1) = 100f'(0).$$

p/w $f'(x) = \frac{100x^{99}}{100} + \frac{99x^{98}}{99} + \dots + \frac{2x}{2} + 1 + 0 \rightarrow f'(0) = 0^{99} + 0^{98} + \dots + 0 + 1$

$$f'(1) = (1)^{99} + (1)^{98} + \dots + (1)^1 + 1$$

$$f'(1) = 100 = 100 \times 1 = 100f'(0)$$

$$f'(0) = 1.$$

QUESTION

Find the derivative of $x^n + ax^{n-1} + a^2x^{n-2} + \dots + a^{n-1}x + a^n$ for some fixed real number a .

$$nx^{n-1} + a(n-1)x^{n-2} + a^2(n-2)x^{n-3} + \dots + a^{n-1}$$

QUESTION

For some constant a and b , find the derivative of

(i) $(x - a)(x - b) = x^2 - (a+b)x + ab \longrightarrow 2x - (a+b)$

(ii) $(ax^2 + b)^2 = a^2x^4 + b^2 + 2abx^2 \longrightarrow 4a^2x^3 + 0 + 4ab$
 $= 4a^2x^3 + 4ab$

(iii) $\frac{x - a}{x - b} = \frac{(x-b) \frac{d(x-a)}{dx} - (x-a) \frac{d(x-b)}{dx}}{(x-b)^2}$
 $= \frac{\cancel{x} - b - \cancel{x} + a}{(x-b)^2} = \frac{a-b}{(x-b)^2}$

QUESTION

Find the derivative of $\frac{x^n - a^n}{x - a}$ for some constant a .

(Sol) Suppose that $f(x) = \frac{x^n - a^n}{x - a}$ be the given function.

Then by Quotient rule, we get

$$\frac{f'(x) = (x - a) \frac{d}{dx}(x^n - a^n) - (x^n - a^n) \frac{d}{dx}(x - a)}{(x - a)^2}$$

$$\Rightarrow f'(x) = \frac{(x - a) \cdot nx^{n-1} - (x^n - a^n)}{(x - a)^2}$$

$$\Rightarrow f'(x) = \frac{nx^n - anx^{n-1} - x^n + a^n}{(x - a)^2}$$

$$\Rightarrow f'(x) = \frac{(n-1)x^n - anx^{n-1} + a^n}{(x - a)^2}$$

Hence the result

QUESTION

Find the derivative of

(i) $2x - \frac{3}{4}$

(ii) $(5x^3 + 3x - 1)(x - 1)$

(iii) $x^{-3}(5 + 3x)$

(iv) $x^5(3 - 6x^{-9})$

(v) $x^{-4}(3 - 4x^{-5})$

(vi) $\frac{2}{x+1} - \frac{x^2}{3x-1}$

$$\left(\frac{2}{x+1}\right)' = \frac{(x+1) \frac{d(2)}{dx} - 2 \frac{d(x+1)}{dx}}{(x+1)^2}$$

$$= \frac{-2}{(x+1)^2}$$

QUESTION



Find the derivative of $\cos x$ from first principle.

#PW

QUESTION

Find the derivative of

$$(i) \quad \sin x \cos x \longrightarrow \sin x \frac{d(\cos x)}{dx} + \cos x \frac{d(\sin x)}{dx} = -\sin^2 x + \cos^2 x = \cos^2 x - \sin^2 x = \cos 2x.$$

$$(ii) \quad \sec x \longrightarrow \sec x \tan x$$

$$(iii) \quad 5 \sec x + 4 \cos x \longrightarrow 5 \sec x \tan x - 4 \sin x$$

$$(iv) \quad \operatorname{cosec} x \longrightarrow -\operatorname{cosec} x \cot x.$$

$$(v) \quad 3 \cot x + 5 \operatorname{cosec} x \longrightarrow -3 \operatorname{cosec}^2 x - 5 \operatorname{cosec} x \cot x.$$

$$(vi) \quad 5 \sin x - 6 \cos x + 7 \longrightarrow 5 \cos x + 6 \sin x$$

$$(vii) \quad 2 \tan x - 7 \sec x \longrightarrow 2 \sec^2 x - 7 \sec x \tan x.$$

Miscellaneous Examples:

Find the derivative of f from the first principle, where f is given by

(i) $f(x) = \frac{2x+3}{x-2}$

(ii) $f(x) = x + \frac{1}{2}$ (HP/W)

(Sol) By first principle, we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{2(x+h)+3}{x+h-2} \right) - \left(\frac{2x+3}{x-2} \right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{(2x+2h+3)(x-2) - (2x+3)(x+h-2)}{h(x+h-2)(x-2)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{2x^2} - \cancel{4x} + \cancel{2hx} - \cancel{4h} + \cancel{3x} - \cancel{6} - \cancel{2x^2} - \cancel{2xh} + \cancel{4x} - \cancel{3x} - \cancel{3h} + \cancel{6}}{h(x+h-2)(x-2)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-7h}{h(x+h-2)(x-2)} = \lim_{h \rightarrow 0} \frac{-7}{(x+h-2)(x-2)} = \frac{-7}{(x+0-2)(x-2)}$$

$$\Rightarrow f'(x) = \frac{-7}{(x-2)^2}$$

Hence the result.

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 12**

QUESTION

Find the derivative of the following functions (it is to be understood that a, b, c, d, p, q, r and s are fixed non-zero constants and m and n are integers):

$$\frac{\sin x + \cos x}{\sin x - \cos x} \quad \text{H/W}$$

Rasib Ayoub Rather



Shadab Gani Bhat



Zaheen Manzoor



Thank You

$$(ax^2+b)^2$$

$$2(ax^2+b) \times \frac{d(ax^2+b)}{dx}$$

$$= 2(ax^2+b) \times 2ax$$

$$= 4ax(ax^2+b)$$

$$(\cos ax)'$$

$$= -\sin ax \times \frac{d(ax)}{dx}$$

$$= -a \sin ax$$